



Clinical and bacteriological effectiveness of three different short-course antibiotic regimens and single-dose fosfomycin for uncomplicated lower urinary tract infections in women (SCOUT): a pragmatic, multicentre, open-label, randomised clinical trial

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Summary

Background Most guidelines recommend nitrofurantoin, fosfomycin, and sometimes pivmecillinam for uncomplicated urinary tract infections (UTIs), but direct comparisons between these antibiotics are needed. This study evaluated the effectiveness and safety of a single dose of fosfomycin compared with two doses of fosfomycin and short-course regimens of nitrofurantoin and pivmecillinam in women with UTI symptoms.

Methods This phase 4, pragmatic, multicentre, parallel-group, open-label, randomised clinical trial was conducted in Spanish primary care centres from 2022 to 2024. Women aged 18 years or older with at least one UTI-specific symptom (dysuria, urinary urgency, urinary frequency, or suprapubic tenderness) and a positive urine dipstick test for either nitrites or leukocyte esterase were randomly assigned (1:1:1) to one of four treatments: a single 3 g dose of fosfomycin, two 3 g doses of fosfomycin, nitrofurantoin (100 mg three times per day for 5 days), or pivmecillinam (400 mg three times per day for 3 days). Doctors and patients were not masked to group assignment. The primary outcome was the proportion of patients with clinical resolution (defined as the disappearance of all infection symptoms) at day 7. This trial is registered with ClinicalTrials.gov (NCT04959331) and EudraCT (2021-001332-26) and is completed.

Findings Of the 804 patients assessed for eligibility between April 4, 2022 and Nov 14, 2024, 768 patients were randomly allocated (191 to the single-dose fosfomycin group, 194 to the two-dose fosfomycin group, 190 to the nitrofurantoin group, and 193 to the pivmecillinam group). Patients had a median age of 48 years (IQR 34–63). Race and ethnicity data were not collected. Among the 720 women with available data included in the primary analysis, single-dose fosfomycin had the lowest proportion of clinical resolution (109 [59%] of 185 patients), while nitrofurantoin had the highest (128 [74%] of 172 patients; difference 15·5 percentage points [95% CI 5·9 to 25·1] vs single-dose fosfomycin; $p=0\cdot0168$), followed by pivmecillinam (127 [70%] of 182; difference 10·9 percentage points [1·1 to 20·6]; $p=0\cdot2352$) and the two-dose fosfomycin group (122 [67%] of 181; difference 8·5 percentage points [−1·4 to 18·3]; $p=0\cdot6935$). Adverse events occurred in 38 (19·9% [95% CI 14·9 to 26·1] of 191 patients who received single-dose fosfomycin, 51 (26·3% [20·6 to 32·9]) of 194 who received two-dose fosfomycin, 51 (26·8% [21·0 to 33·6]) of 190 who received nitrofurantoin, and 41 (21·2% [16·1 to 27·5]) of 193 who received pivmecillinam. Most adverse events were mild and self-limiting, primarily gastrointestinal. Four serious adverse events occurred, of which one was related to the study treatment (one case of pyelonephritis in the pivmecillinam group).

Interpretation Nitrofurantoin was the most effective treatment and single-dose fosfomycin the least effective treatment for UTIs. Adverse events were mild. The role of fosfomycin as a first-line antibiotic for uncomplicated UTI should be re-evaluated.

Funding Carlos III Institute of Health, Spanish Ministry of Science and Innovation, and European Regional Development Fund.

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Introduction

Antibiotics are among the most frequently prescribed medications in primary care, with lower urinary tract

infections (UTIs) being one of the leading indications.¹ The high volume of antibiotic use in this setting contributes to the growing challenge of antimicrobial

Lancet 2026; 407: 1603–13

Published Online

April 20, 2026

[https://doi.org/10.1016/S0140-6736\(25\)02171-3](https://doi.org/10.1016/S0140-6736(25)02171-3)

S0140-6736(25)02171-3

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Research in context

Evidence before this study

We searched PubMed using the terms (randomized OR trial OR systematic review) AND (fosfomycin OR nitrofurantoin OR pivmecillinam) AND (urinary tract infection OR UTI OR cystitis), without language restrictions, for studies published up to Oct 10, 2025. Six systematic reviews were identified that compared the efficacy of single-dose oral fosfomycin with other antibiotic regimens for the treatment of uncomplicated lower urinary tract infections (UTIs) in women. The most recent meta-analysis included ten studies, two of which used two different comparators and two of which compared single doses of other antibiotics. The comparator antibiotics comprised quinolones (five studies), β -lactams or cephalosporins (three studies), nitrofurantoin (three studies), and sulphonamides (one study). This meta-analysis showed comparable proportions of clinical resolution between single-dose fosfomycin and the comparators. One systematic review and meta-analysis, which included three randomised controlled trials, specifically compared fosfomycin and nitrofurantoin and found them to be equivalent in terms of clinical efficacy and safety. Clinical cure rates at both short-term (<4 weeks) and longer-term (>4 weeks) follow-up were similar between groups, and no significant differences were observed in microbiological cure rates within 4 weeks of treatment. However, one open-label randomised clinical trial showed that a 5-day course of nitrofurantoin (100 mg three times per day) was associated with a higher probability of clinical resolution (difference 12% [95% CI 4–21]) and microbiological resolution (difference 11% [95% CI 1–20]) at 28 days post-treatment compared with a single 3 g dose of fosfomycin. The search did not identify any studies comparing the efficacy of different dosing regimens of

fosfomycin with a single-dose regimen in patients diagnosed with uncomplicated UTIs. A Spanish randomised clinical trial published in 2013 found that two doses of 3 g fosfomycin, administered 3 days apart, were as effective as a 3-day course of low-dose ciprofloxacin. The search identified nine studies comparing pivmecillinam at various doses and durations with other antibiotics (two with amoxicillin, three with norfloxacin, and one each with cephalexin, sulfadiazine–trimethoprim, and an aminoglycoside), showing similar efficacy across therapies. No studies were found comparing pivmecillinam with fosfomycin or nitrofurantoin.

Added value of this study

In an open-label, pragmatic, randomised controlled trial involving 768 women presenting to primary care with symptoms of uncomplicated lower UTI, clinical effectiveness at day 7 was lower in those treated with a single 3 g dose of fosfomycin than in those who received other short-course antibiotic regimens (two 3 g doses of fosfomycin, nitrofurantoin at 100 mg three times daily for 5 days, or pivmecillinam at 400 mg three times daily for 3 days).

Implications of all the available evidence

Current guidelines recommend a single 3 g dose of fosfomycin tromethamine or short courses of pivmecillinam or nitrofurantoin as the first-line treatment for uncomplicated UTIs. The results of the current study support findings from a previous randomised clinical trial indicating the lower efficacy of single-dose fosfomycin compared with nitrofurantoin. In light of this new evidence, the use of short-course antibiotic regimens should be encouraged over single-dose fosfomycin.

resistance, emphasising the need for evidence-based prescribing practices that balance efficacy with stewardship.² Up to 10% of women have an acute uncomplicated UTI annually, and over half of women are affected in their lifetime.³ Rising global antibiotic resistance, especially in *Escherichia coli*, is limiting oral treatment options, even for uncomplicated cases.⁴

Current European and US guidelines on the treatment of UTIs recommend the use of either a single 3 g dose of fosfomycin or 100 mg of nitrofurantoin three times per day for 5 days.^{5,6} These recommendations are based on fosfomycin's broad activity against β -lactamase-producing uropathogens and on the low resistance rates to both antibiotics reported in Europe and North America.⁷ Over the past few years, the use of fosfomycin as the preferred therapy for these infections has considerably increased in many European countries, including Spain. However, more than half of Spanish doctors prefer to prescribe two doses of fosfomycin trometamol rather than a single dose, claiming that the single dose is less effective.⁸ Of two systematic reviews comparing single-dose fosfomycin with other antibiotics,

one found fosfomycin to be non-inferior to various comparators, while the other showed no significant differences between fosfomycin and nitrofurantoin.^{9,10} However, a previous randomised trial of 513 women found higher clinical resolution at day 28 with nitrofurantoin (70%) than with a single 3 g dose of fosfomycin (58%).¹¹

Pivmecillinam, a prodrug of mecillinam, has been used for over 40 years in several European countries and in Canada for uncomplicated UTI,^{12,13} and was recently approved by the US Food and Drug Administration for the treatment of uncomplicated UTIs in adult women. Nitrofurantoin, pivmecillinam, and fosfomycin are considered first-line antibiotics for uncomplicated lower UTIs by most guidelines;¹⁴ however, there are no head-to-head trials comparing these three antibiotics.² We aimed to evaluate the clinical effectiveness and safety of a single dose of fosfomycin compared with a two-dose regimen of fosfomycin and short-course regimens of nitrofurantoin and pivmecillinam in women presenting to primary care facilities with symptoms of uncomplicated UTI.

Methods

Study design

This was a phase 4, pragmatic, multicentre, parallel-group, open-label, randomised clinical trial conducted in 34 primary care centres in Spain between April 4, 2022, and Dec 12, 2024. The trial design has been published,¹⁵ and both the trial protocol and statistical analysis plan are included in the appendix (pp 16–72). The study was conducted in four Spanish Autonomous Communities, across which seven different microbiology departments participated. The study was reviewed and approved by the Primary Care Research and Ethics Committee of the Institute for Primary Health Care Research Jordi Gol (IDIAPJGol; Barcelona, Spain; reference number 21/173-AC), and by the Spanish Agency of Medicines and Medical Devices. All patients provided written informed consent. Patients with a history of UTIs were actively involved before the start of the study and contributed to its design. We adhere to the updated 2025 CONSORT guidelines to ensure transparent reporting of our randomised controlled trial, revising all relevant sections accordingly.¹⁶ The trial is registered with ClinicalTrials.gov (NCT04959331) and EudraCT (2021-001332-26).

Participants

Eligible patients were women aged 18 years or older attending primary care consultations with clinical features of uncomplicated community-acquired lower UTIs, including at least one of the four specific symptoms (dysuria, urinary urgency, urinary frequency, or suprapubic tenderness), and no alternative explanation (ie, symptoms suggestive of sexually transmitted infection or vulvovaginitis), and a urine dipstick analysis positive for nitrites or leukocyte esterase.

Exclusion criteria comprised the following: male sex; high suspicion of pyelonephritis, defined as fever of at least 38.5°C or flank pain or tenderness; any condition predisposing the patient to complicated UTI, including indwelling urinary catheter, immunosuppressive therapy, abnormal urinary tract anatomy, severe neurological disease affecting bladder function, or recurrent UTIs (defined as at least three UTIs in the past year or at least two in the past 6 months); pregnancy or breastfeeding; UTI symptoms within the preceding 4 weeks; use of long-term antibiotic prophylaxis; ongoing antibiotic therapy or use of systemic antibiotics within the previous 7 days; symptoms indicative of alternative diagnoses, such as vaginal discharge or pain; known hypersensitivity or allergy to β -lactams, nitrofurantoin, or fosfomycin; moderate to severe chronic renal insufficiency; pre-existing polyneuropathy; history of pulmonary or hepatic reactions, or peripheral neuropathy following nitrofurantoin use; glucose-6-phosphate dehydrogenase deficiency; porphyria, systemic primary carnitine deficiency, or organic acidurias such as methylmalonic aciduria or propionic acidemia; oesophageal stricture; concurrent use of allopurinol (due to increased risk of allergic skin reactions with mecillinam),

probenecid (which reduces renal excretion of mecillinam), or valproate; active malignancy or terminal illness; current participation in another randomised clinical trial; previous enrolment in the present study; residency in long-term care institutions; or anticipated difficulties in attending scheduled follow-up visits.

Randomisation and masking

This was an open-label trial; neither physicians nor patients were masked to the assignment of the patient to the study group. Participants were randomly assigned to one of four treatment groups (1:1:1:1): a single-dose fosfomycin group, a two-dose fosfomycin group, a nitrofurantoin group, and a pivmecillinam group. Block randomisation was used (block size eight), with stratification by study centre. Randomisation was independently executed at each participating site, and investigators were masked to the randomisation process.

Randomisation was done before receipt of baseline urine culture results. Patient allocation was implemented through Research Electronic Data Capture (REDCap), a secure, web-based application designed for data collection and randomisation management.¹⁷ The system automatically assigned the next available treatment from the preloaded randomisation list once participant eligibility was confirmed. Access to the randomisation module within REDCap was strictly limited to the site investigator teams. The independent statistician who generated the randomisation list had no access to any patient-level data throughout the study.

Procedures

After randomisation, participants were informed of their assigned treatment strategy and instructed on appropriate actions in case of symptom worsening or lack of improvement. They were provided with the study medication and given guidance on its proper administration: for the single-dose fosfomycin group, a single oral 3 g dose of fosfomycin trometamol; for the two-dose fosfomycin group, two oral 3 g doses of fosfomycin trometamol, with the second dose taken 24 h after the first dose; for the nitrofurantoin group, oral nitrofurantoin at a dosage of 100 mg three times per day for 5 days; and for the pivmecillinam group, oral pivmecillinam at a dosage of 400 mg three times per day for 3 days. Participants allocated to the fosfomycin groups were instructed to take the medication on an empty stomach or at least 2 h after meals. Participants in the nitrofurantoin and pivmecillinam groups were recommended to take the doses with meals. Each patient was provided with a symptom diary to document the progression of their symptoms from days 0 to 7, along with instructions for its completion and a request to return it on day 14. Follow-up visits were scheduled on days 7, 14, and 28, and could be performed within 2 days following each of these timepoints (eg, days 7–9 for the day 7 visit).

See Online for appendix

Urine cultures were obtained at baseline and during day 14 and day 28 visits. A midstream clean-catch urine collection was recommended. In each of the primary care centres, urine specimens were delivered to the microbiology laboratory within 24 h of collection. All tubes contained boric acid. If processing (from collection to plating) was not possible within 2 h, the sample was stored at 4°C until the following morning, at which point it was sent to the microbiology department. All urine samples were processed according to routine laboratory procedures, and susceptibility to common antibiotics (fosfomycin and nitrofurantoin) was evaluated by analysing the breakpoints based on minimum inhibitory concentrations (MICs) or zone diameter breakpoints. All study antibiotics were assessed for susceptibility using zone diameter breakpoints in accordance with the European Committee on Antimicrobial Susceptibility Testing guidelines.¹⁸ Susceptibility to pivmecillinam was assessed by the disk diffusion method, whereas susceptibility to the other antibiotics was assessed by automated broth microdilution. In the presence of substantial bacteriuria (ie, $\geq 10^3$ colony-forming units [CFU] per mL of a single pathogen, according to current European guidelines for women with symptoms of lower UTI),⁵ the isolates were examined for resistance mechanisms and patterns and MICs to common antibiotics, including fosfomycin, nitrofurantoin, and pivmecillinam.

Outcomes

The primary outcome was clinical effectiveness on day 7, measured by clinical resolution, which was defined as the disappearance of all infection symptoms, as reported by patients, on day 7, indicating the day the infection was considered cured. Patient outcomes were classified as either clinical resolution, improvement, or failure. Improvement was defined as symptom relief with some persistence, but without signs of infection and with no need for further antibiotics. Failure was defined as a change or addition of antibiotics due to lack of efficacy or ongoing infection.

Several secondary outcomes were considered: time (in days) until patient-reported clinical resolution; bacteriological eradication measured at day 14, defined as eradication of the infecting strain with no recurrence of bacteriuria ($<10^3$ CFU/mL); bacteriological eradication at day 28; clinical resolution at day 14; clinical resolution at day 28; distribution of microorganisms causing UTI and resistance rates of uropathogens to the study antibiotics; number of patients who used antibiotics other than the study medication during the trial; recurrences, defined as the reappearance of infection within the first 4 weeks after clinical resolution of the initial episode; patient satisfaction within each treatment strategy; visual analogue scale scores to assess the general health status (0–100 scale); and patient adherence to the study medication in each strategy, considering patients to be adherent if they reported taking at least 80% of the prescribed doses.

Adverse events were specifically assessed at contact points on days 7, 14, and 28, with patients also able to report any harms at any time during the 28-day follow-up period, with a focus on pyelonephritis as a special event of interest. Adverse events were evaluated by a clinician from the recruitment team, who was therefore aware of the treatment group to which the patient had been assigned. Events were categorised by severity, seriousness, and relatedness to the study drug. The investigator was responsible for reporting all events deemed serious, and the causal relationship with the investigational product was determined based on clinical judgement. In the case of a serious adverse event, the pharmacovigilance department was notified, and a serious adverse event notification form from the Spanish Agency of Medicines and Medical Devices was completed.

Two additional secondary outcomes will be reported separately: the predictive value of clinically confirmed UTI, based on the different clinical criteria collected in the study, for the diagnosis of microbiologically confirmed UTI; and the cost-effectiveness of each trial strategy.

Statistical analysis

We hypothesised a clinically important minimum difference of 10 percentage points in the primary outcome of clinical resolution, based on the latest guideline on the evaluation of medicinal products indicated for the treatment of bacterial infections, issued by the European Medicines Agency (EMA).¹⁹ Assuming a clinical efficacy of 75% for single-dose fosfomycin, as recently found in a systematic review,¹⁰ with a two-sided type I error rate of 5%, and a statistical power of 80% ($\beta=0.2$), at least 253 patients per group with an available primary outcome were required for the study to be conclusive. Considering an estimated 10% dropout rate in each study group, we aimed to recruit 280 patients per group, for a total planned inclusion of 1120 patients.

The primary statistical comparison of the primary outcome was a two-sided χ^2 test comparing the three short-course antibiotic regimens with the single dose of fosfomycin. 95% CIs were estimated using normal approximation. Pairwise comparisons were adjusted using the Bonferroni method.²⁰ Efficacy analyses were primarily based on the intention-to-treat (ITT) population, defined as all patients who were randomly assigned to a treatment group and received at least one dose of the assigned treatment. Per-protocol analysis of the primary outcome was used as a secondary analysis; the per-protocol population was defined as all women in the ITT population who completed all efficacy assessments and had no protocol deviations, and who were adherent to the medication (taking at least 80% of the prescribed doses). Missing data were minimised through trial monitoring and cross-checking. For the main outcome, missingness was assumed to be not at random. Clinical outcomes at days 14 and 28 were imputed using the last observation carried forward from day 7. Secondary

microbiological outcomes were analysed using a complete-case approach.

We did two sensitivity analyses of the primary outcome. In the first, we adjusted the main analysis for the stratification variable of baseline urine culture results. In the second, missing data for the primary outcome were imputed using a best-case scenario (all missing values imputed as recovered), worst-case scenario (all missing values imputed as treatment failure), extreme superiority scenario (missing values imputed as treatment failure in

the single-dose fosfomycin group and as recovered in the two-dose fosfomycin, nitrofurantoin, and pivmecillinam groups), and counterintuitive scenario (missing values imputed as recovered in the single-dose fosfomycin group and as treatment failures in the two-dose fosfomycin, nitrofurantoin, and pivmecillinam groups). Additionally, efficacy outcomes were reanalysed for individuals who tested positive for *E coli* in baseline urine cultures, as well as separately for premenopausal and postmenopausal women.

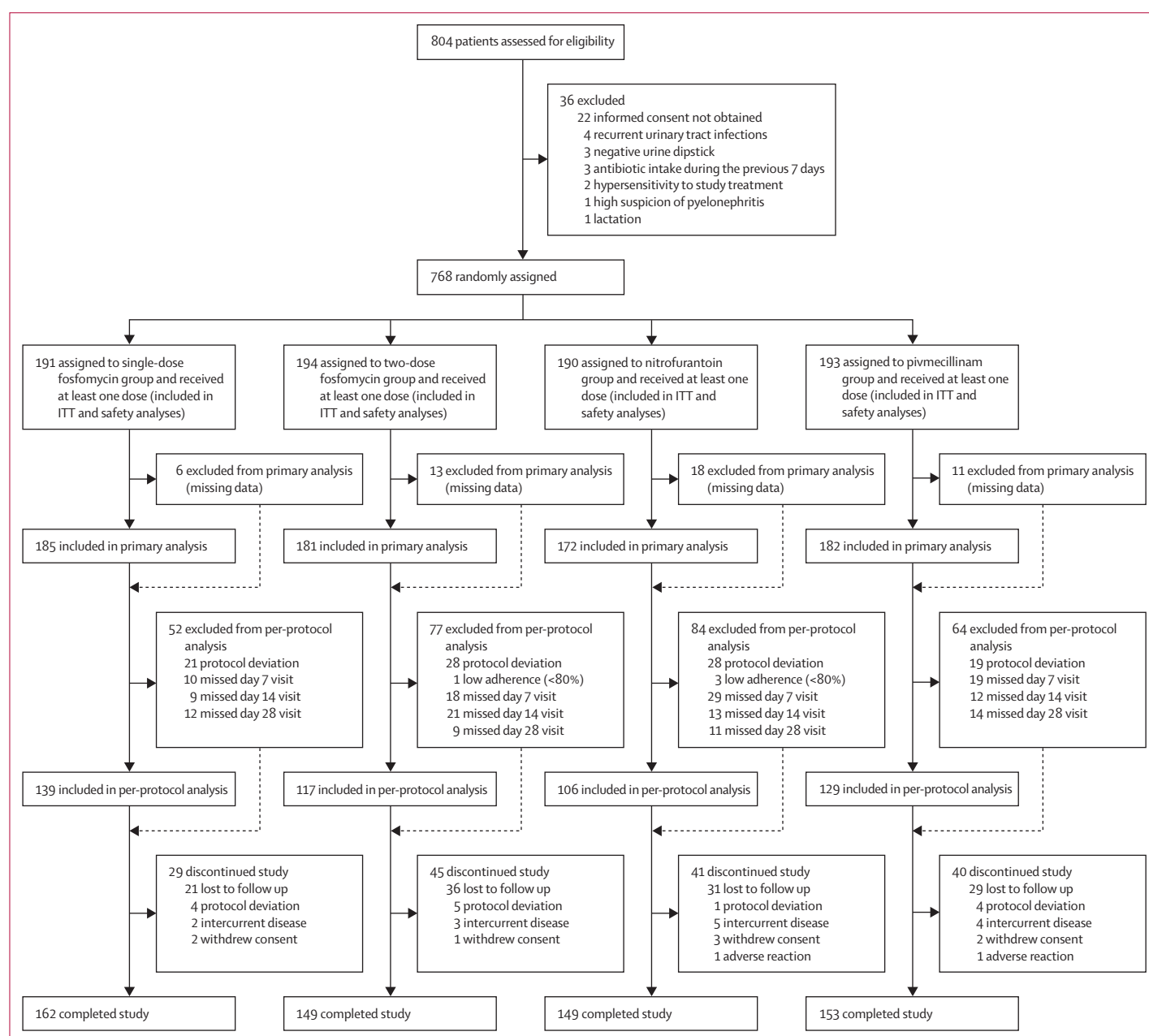


Figure: Trial profile
ITT=intention-to-treat.

	Single-dose fosfomycin group (n=191)	Two-dose fosfomycin group (n=194)	Nitrofurantoin group (n=190)	Pivmecillinam group (n=193)
Median age, years	50 (35·5–64·5)	47·5 (34–62)	47 (33–62)	49 (34–64)
Previous UTI				
At any time	164 (86%)	161 (83%)	152 (80%)	154 (80%)
In the year preceding inclusion in the trial	83/164 (51%)	75/161 (47%)	71/152 (47%)	76/154 (49%)
General health condition, median visual analogue scale*	70 (50–90)	70 (50–90)	70 (50–90)	70 (57·5–80)
Median days since onset of symptoms	2 (1–3)	2 (1–4)	2 (1–4)	2 (1–3)
UTI-specific symptoms				
Dysuria	163 (85%)	174 (90%)	154 (81%)	162 (84%)
Urinary urgency	161 (84%)	167 (86%)	153 (81%)	166 (86%)
Urinary frequency	165 (86%)	172 (89%)	166 (87%)	178 (92%)
Suprapubic pain	141 (74%)	137 (71%)	127 (67%)	146 (76%)
Median number of UTI-specific symptoms†	4 (3–4)	4 (3–4)	3 (3–4)	4 (3–4)
Other symptoms				
Feverish sensation	27 (14%)	19 (10%)	21 (11%)	27 (14%)
General discomfort	91 (48%)	95 (49%)	85 (45%)	89 (46%)
Restriction of daily activities	57 (30%)	44 (23%)	44 (23%)	48 (25%)
Haematuria	67 (35%)	77 (40%)	69 (36%)	79 (41%)
Cloudy urine	89 (47%)	99 (51%)	91 (48%)	103 (53%)
Foul-smelling urine	55 (29%)	60 (31%)	45 (24%)	41 (21%)
Urinalysis				
Urine dipstick positive for leukocytes	189/191 (99%)	189/193 (98%)	184/190 (97%)	186/193 (96%)
Urine dipstick positive for nitrites	57/191 (30%)	48/193 (25%)	49/190 (26%)	46/192 (24%)
Baseline urine culture obtained	187 (98%)	185 (95%)	183 (96%)	183 (95%)
Confirmed UTI‡	109/187 (58%)	102/185 (55%)	91/183 (50%)	119/183 (65%)
Contaminated	24/187 (13%)	20/185 (11%)	22/183 (12%)	19/183 (10%)
Negative	54/187 (29%)	63/185 (34%)	70/183 (38%)	45/183 (25%)
Number of uropathogens isolated‡	107	104	95	120
<i>Escherichia coli</i>	74/107 (69%)	78/104 (75%)	67/95 (71%)	88/120 (73%)
<i>Staphylococcus saprophyticus</i>	9/107 (8%)	5/104 (5%)	6/95 (6%)	10/120 (8%)
<i>Proteus mirabilis</i>	9/107 (8%)	5/104 (5%)	6/95 (6%)	4/120 (3%)
<i>Klebsiella pneumoniae</i>	7/107 (7%)	4/104 (4%)	4/95 (4%)	8/120 (7%)
Others§	8/107 (7%)	12/104 (12%)	12/95 (13%)	10/120 (8%)
Proportion of strains resistant to the allocated antibiotic¶	10/100 (10%)	10/96 (10%)	9/87 (10%)	12/76 (16%)
Proportion of <i>E coli</i> strains resistant to the allocated antibiotic	1/74 (1%)	2/78 (3%)	0/65	5/60 (8%)

Additional information regarding patient characteristics is provided in the appendix (p 2). UTI=urinary tract infection. *Visual analogue scale ranged from 0 (worst health status) to 100 (best health status). †UTI-specific symptoms included are dysuria, urinary urgency, urinary frequency, and suprapubic pain. ‡Positive culture was defined as the growth of at least 10³ colony-forming units per mL of at least one uropathogen; some cultures had polymicrobial growth; there were two patients with missing details on the uropathogen identified in the single-dose fosfomycin group. §*Citrobacter koseri*, *Enterococcus faecalis*, *Streptococcus agalactiae*, *Klebsiella aerogenes*, *Morganella morganii*, *Lactobacillus crispatus*, *Staphylococcus saccharolyticus*, *Citrobacter freundii*, *Staphylococcus aureus*, *Pseudomonas aeruginosa*, *Salmonella typhimurium*, *Enterobacter cloacae*, or *Klebsiella variicola*. ¶Antibiograms for study treatments were done in cultures positives for any uropathogen; a pivmecillinam antibiogram was an additional determination compared with usual practice (this extra determination was not done on 43 occasions); a nitrofurantoin antibiogram was not done on four occasions; a fosfomycin antibiogram was not done on nine occasions in the single-dose fosfomycin group and on six occasions in the two-dose fosfomycin group; all these cases were treated as missing values. ||Antibiograms for study treatments were done in cultures positive for *E coli*; a pivmecillinam antibiogram was not done on 28 occasions; a nitrofurantoin antibiogram was not done on two occasions; all these cases were treated as missing values.

Table 1: Baseline characteristics in the intention-to-treat population (n=768)

The safety population included all patients who received any trial drug. Safety data were summarised using descriptive statistics. Adverse events were compared across three short-course antibiotic regimens and single-dose fosfomycin using 95% CIs (Wilson method). Risk differences and relative risks (with 95% CIs) were calculated for each short course versus single-dose fosfomycin.

Two-sided Bonferroni-adjusted p values <0·05 were considered statistically significant. Analyses were done with R software (version 4.3.1).

Role of the funding source

The funders of the study had no role in study design, data collection, data analysis, data interpretation, or writing of the report.

	Clinical resolution on day 7	Clinical resolution on day 7 in patients with a positive baseline urine culture	Clinical resolution on day 14	Clinical resolution on day 28	Bacterial eradication on day 14	Bacterial eradication on day 28
Two-dose fosfomycin vs single-dose fosfomycin	8.5 (−1.4 to 18.3)	13.7 (0.3 to 27.2)	5.4 (−3.9 to 18.3)	5.3 (−4.1 to 14.8)	4.0 (−10.5 to 18.4)	4.6 (−10.7 to 19.8)
Nitrofurantoin vs single-dose fosfomycin	15.5 (5.9 to 25.1)*	34.9 (22.9 to 46.8)*	12.7 (3.8 to 21.6)*	12.6 (3.5 to 21.7)*	−3.0 (−18.4 to 12.4)	8.3 (−6.7 to 23.3)
Pivmecillinam vs single-dose fosfomycin	10.9 (1.1 to 20.6)	22.0 (9.6 to 34.5)*	8.8 (−0.3 to 17.9)	8.8 (−0.5 to 18.0)	−0.1 (−14.5 to 14.3)	−0.2 (−15.5 to 15.1)

Values are percentage points difference (95% CI). Numbers and percentages of participants with each outcome per group are provided in the appendix (p 5). *Significant difference at the 5% level after adjustment using the Bonferroni method.

Table 2: Pairwise comparisons of efficacy outcomes

Results

Between April 4, 2022, and Nov 14, 2024, 804 patients were assessed for eligibility, of whom 36 were excluded. The remaining 768 patients were randomly allocated to one of the four study groups (191 to the single-dose fosfomycin group, 194 to the two-dose fosfomycin group, 190 to the nitrofurantoin group and 193 to the pivmecillinam group) and received at least one dose of study treatment and constituted the ITT population (figure). The target sample size was not reached, as the study was terminated due to the exhaustion of available funding. Within the ITT population, 491 (64%) completed the study according to the protocol and comprised the per-protocol population.

The median age in the ITT population was 48 years (IQR 34–63). Baseline characteristics were balanced among the groups (table 1; appendix p 2). The baseline characteristics of patients who completed the study versus those who discontinued are described in the appendix (p 3). 631 (82%) of the population had previously had a UTI at any time, with 305 (48%) having had a UTI in the previous year. Urinary frequency was the most predominant symptom (681 [89%] patients), followed by dysuria (653 [85%]) and urinary urgency (647 [84%]).

Baseline urine culture results were available for most patients (738 [96%]), among whom 421 (57%) tested positive for at least one uropathogen. The proportion was higher among women presenting with three or four UTI-specific symptoms (appendix p 4). The most frequently isolated microorganism was *E coli* (307 [72%] of 426 isolates; table 1). 41 patients presented with a uropathogen resistant to the allocated study treatment, most often due to microorganisms other than *E coli*. The distribution of these patients was similar across study groups, with a slightly higher incidence in the pivmecillinam group.

Among 720 patients with available primary outcome data, nitrofurantoin was the most effective treatment at day 7, with clinical resolution occurring in 128 (74%) of 172 patients, followed by pivmecillinam (127 [70%] of 182) and the two-dose fosfomycin group (122 [67%] of 181; appendix p 5). The single-dose fosfomycin regimen was the least effective (109 [59%] of 185). Pairwise comparisons of the proportion of patients with clinical resolution at day 7 between study groups showed a clinically meaningful

	All patients in ITT population	Population with a positive baseline urine culture
Best-case scenario*		
Two-dose fosfomycin vs single-dose fosfomycin	9.4 (−0.1 to 18.9)	14.4 (1.4 to 27.5)
Nitrofurantoin vs single-dose fosfomycin	16.6 (7.5 to 25.8)†	34.7 (23.2 to 46.2)†
Pivmecillinam vs single-dose fosfomycin	11.3 (1.9 to 20.7)	21.6 (9.4 to 33.8)†
Worst-case scenario‡		
Two-dose fosfomycin vs single-dose fosfomycin	5.8 (−4.0 to 15.6)	11.3 (−2.0 to 24.6)
Nitrofurantoin vs single-dose fosfomycin	10.3 (0.6 to 20.0)	28.7 (16.1 to 41.2)†
Pivmecillinam vs single-dose fosfomycin	8.7 (−1.0 to 18.4)	21.0 (8.6 to 33.4)†
Extreme superiority§		
Two-dose fosfomycin vs single-dose fosfomycin	12.5 (3.0 to 22.1)†	17.2 (4.1 to 30.2)
Nitrofurantoin vs single-dose fosfomycin	19.8 (10.5 to 29.0)†	37.5 (25.9 to 49.0)†
Pivmecillinam vs single-dose fosfomycin	14.4 (5.0 to 23.9)†	24.3 (12.1 to 36.5)†
Counterintuitive outcomes¶		
Two-dose fosfomycin vs single-dose fosfomycin	2.7 (−7.0 to 12.4)	8.6 (−4.7 to 21.8)
Nitrofurantoin vs single-dose fosfomycin	7.2 (−2.5 to 16.8)	25.9 (13.4 to 38.5)†
Pivmecillinam vs single-dose fosfomycin	5.6 (−4.0 to 15.2)	18.2 (5.8 to 30.6)†

Data are percentage points difference (95% CI) in proportion of patients with clinical resolution at day 7. ITT=intention-to-treat. *All missing values for the primary outcome are imputed as recovered. †Significant difference at the 5% level after adjustment using the Bonferroni method. ‡All missing values for the primary outcome are imputed as treatment failures. §For the primary outcome, missing values in the single-dose fosfomycin group are imputed as treatment failures, whereas in the two-dose fosfomycin, nitrofurantoin, and pivmecillinam groups, they are imputed as recovered. ¶For the primary outcome, missing values in the single-dose fosfomycin group are imputed as recovered, whereas in the two-dose fosfomycin, nitrofurantoin, and pivmecillinam groups, they are imputed as treatment failures.

Table 3: Pairwise main outcome comparisons with missing data imputation

difference greater than 10 percentage points in the nitrofurantoin group versus the single-dose fosfomycin group (15.5 percentage points [95% CI 5.9–25.1]; $p=0.0168$) and in the pivmecillinam group versus the single-dose fosfomycin group (10.9 percentage points [1.1–20.6]; $p=0.2352$; table 2). The difference in the proportion of patients with clinical resolution between the two-dose fosfomycin group and the single-dose fosfomycin group was 8.5 percentage points (−1.4 to 18.3; $p=0.6935$). Similar results were obtained in the per-protocol analysis (appendix p 6). In a sensitivity analysis of patients with a microbiologically confirmed UTI at baseline, compared with the single-dose fosfomycin group, clinical resolution at day 7 was statistically significantly higher in the nitrofurantoin group (34.9 percentage points

[22.9–46.8]; $p < 0.0001$) and pivmecillinam group (22.0 percentage points [9.6–34.5]; $p = 0.0068$), and a clinically relevant difference of 13.7 percentage points (0.3–27.2; $p = 0.3986$) was found in the two-dose fosfomycin group (table 2; appendix p 7). No differences between groups were detected for patients who did not test positive for baseline urine cultures (appendix p 7). Additional sensitivity analyses of the primary outcome based on imputation of missing data (worst-case, best-case, extreme superiority, and counterintuitive scenarios) showed consistent results across all scenarios, with ranking from highest to lowest clinical resolution maintaining the order nitrofurantoin, pivmecillinam, two-dose fosfomycin, and single-dose fosfomycin (table 3). Patients in the single-dose fosfomycin group showed the lowest rate of clinical resolution and required more time to reach clinical resolution (appendix pp 8–9).

Nitrofurantoin also had the highest clinical resolution at days 14 and 28, occurring in 139 (81%) and 136 (79%) of 172 patients, respectively, whereas the single-dose fosfomycin regimen was the least effective, with rates of 126 (68%) and 123 (66%) of 185 patients, respectively (appendix p 5). Both microbiological response and

bacteriological eradication were similar between groups at days 14 and 28 (table 2; appendix p 5). A subanalysis of patients with *E coli*-positive cultures at baseline showed nitrofurantoin had the highest clinical efficacy, with significantly higher proportions of patients with clinical resolution than other treatments. Microbiological responses at day 14 were similar; however, nitrofurantoin showed relatively better outcomes at day 28 (appendix p 10).

The resistance rates of uropathogens to the allocated study group antibiotics were similar, ranging from around 10% for the single-dose fosfomycin, two-dose fosfomycin, and nitrofurantoin regimens, to 16% for pivmecillinam (table 1). For *E coli* in particular, the differences in the percentage of resistance among groups ranged from 0% for nitrofurantoin to 8% for pivmecillinam (table 1). No differences were observed in the efficacy outcomes between postmenopausal and premenopausal women (appendix p 11). An additional antibiotic course was required in 57 (30%) of 191 patients in the single-dose fosfomycin group, 37 (19%) of 194 in the two-dose fosfomycin group, 30 (16%) of 190 in the nitrofurantoin group, and 31 (16%) of 193 in the pivmecillinam group.

	Single-dose fosfomycin group (n=191)	Two-dose fosfomycin group (n=194)	Nitrofurantoin group (n=190)	Pivmecillinam group (n=193)
Patients with one or more adverse events	38/191 (19.9% [14.9–26.1])	51/194 (26.3% [20.6–32.9])	51/190 (26.8% [21.0–33.6])	41/193 (21.2% [16.1–27.5])
Discontinuation due to adverse event	0/191 (0.0% [0.0–2.0])	1/194 (0.5% [0.1–2.9])	5/190 (2.6% [1.1–6.0])	3/193 (1.6% [0.5–4.5])
Patients with at least one serious adverse event*	0/191 (0.0% [0.0–2.0])	1/194 (0.5% [0.1–2.9])	1/190 (0.5% [0.1–2.9])	2/193 (1.0% [0.3–3.7])
Related to trial drug†	0/191 (0.0% [0.0–2.0])	0/194 (0.0% [0.0–1.9])	0/190 (0.0% [0.0–2.0])	1/193 (0.5% [0.1–2.9])
Total number of adverse events reported	41	73	77	57
Expected adverse events‡	33/41 (80.5% [66.0–89.8])	55/73 (75.3% [64.4–83.8])	54/77 (70.1% [59.2–79.2])	38/57 (66.7% [53.7–77.5])
Adverse event of special interest: pyelonephritis	0/191 (0.0% [0.0–2.0])	0/194 (0.0% [0.0–1.9])	2/190 (1.1% [0.3–3.8])	1/193 (0.5% [0.1–2.9])
Adverse event severity§				
Mild	30/40 (75.0% [59.8–85.8])	60/72 (83.3% [73.1–90.2])	55/77 (71.4% [60.5–80.3])	40/54 (74.1% [61.1–83.9])
Moderate	9/40 (22.5% [12.3–37.5])	11/72 (15.3% [8.8–25.3])	21/77 (27.3% [18.6–38.1])	10/54 (18.5% [10.4–30.8])
Severe	1/40 (2.5% [0.4–12.9])	1/72 (1.4% [0.2–7.5])	1/77 (1.3% [0.2–7.0])	4/54 (7.4% [2.9–17.6])
Most common adverse events¶				
Diarrhoea	17/191 (8.9% [5.6–13.8])	27/194 (13.9% [9.7–19.5])	7/190 (3.7% [1.8–7.4])	7/193 (3.6% [1.8–7.3])
Abdominal pain	3/191 (1.6% [0.5–4.5])	6/194 (3.1% [1.4–6.6])	7/190 (3.7% [1.8–7.4])	5/193 (2.6% [1.1–5.9])
Vulvovaginal candidiasis	6/191 (3.1% [1.4–6.7])	3/194 (1.5% [0.5–4.4])	3/190 (1.6% [0.5–4.5])	5/193 (2.6% [1.1–5.9])
Nausea	2/191 (1.0% [0.3–3.7])	1/194 (0.5% [0.1–2.9])	6/190 (3.2% [1.5–6.7])	6/193 (3.1% [1.4–6.6])
Headache	0/191 (0.0% [0.0–2.0])	2/194 (1.0% [0.3–3.7])	5/190 (2.6% [1.1–6.0])	7/193 (3.6% [1.8–7.3])
Fatigue	0/191 (0.0% [0.0–2.0])	2/194 (1.0% [0.3–3.7])	9/190 (4.7% [2.5–8.8])	2/193 (1.0% [0.3–3.7])
Vaginitis	2/191 (1.0% [0.3–3.7])	3/194 (1.5% [0.5–4.4])	2/190 (1.1% [0.3–3.8])	1/193 (0.5% [0.1–2.9])
Pruritus	0/191 (0.0% [0.0–2.0])	3/194 (1.5% [0.5–4.4])	2/190 (1.1% [0.3–3.8])	2/193 (1.0% [0.3–3.7])
Dizziness	1/191 (0.5% [0.1–2.9])	0/194 (0.0% [0.0–1.9])	4/190 (2.1% [0.8–5.3])	1/193 (0.5% [0.1–2.9])
Somnolence	0/191 (0.0% [0.0–2.0])	0/194 (0.0% [0.0–1.9])	5/190 (2.6% [1.1–6.0])	0/193 (0.0% [0.0–2.0])

Data are n/N (% [95% CI]); values represent the number of affected patients, except where otherwise noted. The safety population included all the patients who had received any dose of a trial drug. Events were coded according to the terms used in the Medical Dictionary for Regulatory Activities (version 27.1). *Serious adverse events recorded were acute myocardial infarction (pivmecillinam group), mesenteric panniculitis (two-dose fosfomycin group), and pyelonephritis (nitrofurantoin and pivmecillinam groups), all of which resolved with recovery. †The relationship of the severe adverse event to a trial drug was assessed by the pharmacovigilance department; one case of pyelonephritis in the pivmecillinam group was judged to be related to the study drug. ‡Expected adverse events were those described in the technical data sheet for each study treatment; values represent the number of expected adverse events out of all adverse events reported. §Values represent the number of adverse events of each severity level out of all adverse events reported for which severity data were available; severity data were missing for five adverse events. ¶Adverse events reported during treatment are listed in the order of descending overall frequency; events that occurred in fewer than five cases are not included in the table; the maximum number of adverse events reported per patient was four (occurring in four patients).

Table 4: Summary of adverse events (safety population)

When this exposure was adjusted based on the baseline urine culture result, the differences increased in patients with a positive result, with pivmecillinam being the treatment that required the fewest additional courses (appendix p 12). The proportions of patients with recurrence were similar across study groups, although slightly higher in the fosfomycin groups. 88% of all reconsultation visits (ranging from 83% in the nitrofurantoin group to 96% in the two-dose fosfomycin group) were in primary care settings (appendix p 13). Of the 681 patients with available adherence information, good adherence was observed in all 179 (100%) patients taking single-dose fosfomycin, 168 (98%) of 171 taking two-dose fosfomycin, 150 (94%) of 160 taking nitrofurantoin, and 171 (99%) of 173 taking pivmecillinam. Satisfaction was similar across groups (appendix p 13).

Adverse events occurred in 38 participants (19.9% [95% CI 14.9–26.1]) in the single-dose fosfomycin group, 51 (26.3% [20.6–32.9]) in the two-dose fosfomycin group, 51 (26.8% [21.0–33.6]) in the nitrofurantoin group, and 41 (21.2% [16.1–27.5]) in the pivmecillinam group (table 4). Compared with single-dose fosfomycin, the risk difference for the occurrence of any adverse event was 6.4 percentage points (95% CI –5.5 to 18.0) for two-dose fosfomycin, 6.9 (–5.1 to 18.7) for nitrofurantoin, and 1.3 (–10.1 to 12.7) for pivmecillinam. The corresponding relative risks were 1.3 (95% CI 0.9–1.9), 1.4 (0.9–2.0), and 1.1 (0.7–1.6), respectively (appendix p 14). Most adverse events were mild (75%, 83%, 71%, and 74% of the adverse events in each group, respectively), and were consistent with the drugs' known safety profiles (table 4). Moderate and severe events occurred in less than 21% and less than 3% of patients, respectively. Treatment discontinuations due to adverse events were infrequent, ranging from 0.0% (0.0–2.0) in the single-dose fosfomycin group to 2.6% (1.1–6.0) in the nitrofurantoin group.

A total of four serious adverse events occurred during the trial, only one of which was considered related to the study drug. Two serious adverse events occurred in the pivmecillinam group (one acute myocardial infarction and one pyelonephritis [related to study drug]), one in the nitrofurantoin group (pyelonephritis), and one in the two-dose fosfomycin group (mesenteric panniculitis). All serious adverse events resolved completely, with full patient recovery.

The most common adverse events were diarrhoea, occurring particularly among patients allocated to fosfomycin, followed by abdominal pain and vulvovaginal candidiasis. Pyelonephritis, a prespecified event of special interest, occurred in two patients treated with nitrofurantoin (1.1% [95% CI 0.3–3.8]) and one treated with pivmecillinam (0.5% [0.1–2.9]), but not among those receiving fosfomycin. Overall, no unexpected safety signals were identified, and the incidence and nature of adverse events were similar across treatments and consistent with known drug safety profiles.

Discussion

In women with uncomplicated UTIs, single-dose fosfomycin was less effective by day 7 than other short-course regimens, with clinical resolution occurring in a significantly lower proportion of patients with microbiologically confirmed UTIs at baseline—34.9 percentage points lower versus the nitrofurantoin group and 22.0 percentage points lower versus the pivmecillinam group. Our results support findings from a previous trial that showed nitrofurantoin to be substantially more effective than single-dose fosfomycin.¹¹ Our study is novel in that it compares four antibiotic regimens head to head, and it is the first to compare two different formulations of fosfomycin. Furthermore, no studies have previously compared pivmecillinam with either of the other two antibiotics.

We conducted a pragmatic clinical trial, with clinical resolution on day 7 selected as the primary outcome, reflecting standard practice in primary care. In women with uncomplicated UTIs, clinical resolution typically occurs within 1 week.²¹ A Swedish study reported median recovery times of 4 days (IQR 2–5) for women treated with antibiotics and 6.5 days (3–10) for those not treated.²² Clinical outcomes are most relevant to patients and guide follow-up decisions, such as additional visits or treatments. In primary care, uncomplicated UTIs are typically managed without microbiological analysis. Including microbiological response in a composite primary outcome, as recommended by the EMA¹⁹ and considered in recent trials evaluating new drugs for uncomplicated UTI,²³ would have excluded patients without a positive baseline culture (317 [43%] of 738 in our study, including those with negative or contaminated cultures). In a study by Huttner and colleagues,¹¹ which compared nitrofurantoin with a single dose of fosfomycin using the same inclusion criteria as in our study (one or more of the localising urinary tract symptoms, as proposed by Nicolle),³ 72% of urine cultures were positive, although the study population was younger (median age 44 years [31–64]).¹¹ Age is associated with UTI misdiagnosis, suggesting that some of our participants might have had non-infectious urethral symptoms at enrolment. Other primary care studies report even lower rates of microbiologically confirmed UTIs. For instance, a multicentre trial conducted across four European countries, which also used the same inclusion criteria, found that only 36% of nearly 800 women with suspected UTIs had culture-confirmed infections.²⁴

Accurately diagnosing UTIs in primary care remains challenging. Current methods—based on symptoms, signs, and dipstick tests—lack precision, leading to both overtreatment and undertreatment. Improved point-of-care diagnostics are urgently needed, as existing tools such as urine dipsticks have low sensitivity and specificity.²⁵ Additionally, these tools cannot provide information on the antibiotic susceptibility of pathogens. Bacterial culture tests on solid media take 18–24 h, delaying clinical decisions, and a previous trial showed minimal effects on

improving antibiotic use.²⁶ The gold standard—microscopy, culture, and susceptibility testing—takes 2–3 days and can be limited by contamination by host flora. Although our trial's contamination rate was under 13%, other studies report rates up to 30%.²⁷ Delays and sample issues can lead to inconclusive results, highlighting the urgent need for faster, more accurate point-of-care tests to guide antibiotic prescribing in primary care.

The resistance of *E coli* to fosfomycin is primarily attributed to plasmid-mediated genes encoding fosfomycin-modifying enzymes, which raises concerns about the potential for broader resistance development. Although there has been a notable increase in fosfomycin consumption over the past decade in Spain, resistance rates remain below 10%, as also observed in this study. Data from more than 70 000 community urine cultures in Catalonia in 2024 showed that 3·8% of *E coli* isolates were resistant to fosfomycin, with resistance to nitrofurantoin remaining under 1%.²⁸ The discrepancy between the reduced in vivo fosfomycin efficacy and what would be expected based on in-vitro resistance rates suggests microbiological and pharmacological factors.²⁹ A single 3 g dose might not sustain urine concentrations above the MIC for a sufficient duration, particularly given interindividual variability in pharmacokinetics. The improved outcomes observed with two-dose regimens suggest that single dosing might be insufficient to maintain effective exposure, consistent with fosfomycin's predominantly time-dependent antibacterial activity.³⁰ The reduced activity of fosfomycin in low urine pH and *E coli* persistence in the bladder could explain its lower efficacy, as a single dose might not sustain a MIC in the urine for long enough to clear the infection.³¹ Future studies will be required to investigate these hypotheses.

The planned sample size was not reached in our study, representing a key limitation. For the primary outcome, the final sample size yielded an estimated power of 66·7%, below the originally intended 80%. Although statistically significant results were observed for the primary and some secondary outcomes, the reduced sample size might have limited the detection of clinically relevant differences (ie, those ≥ 10 percentage points) due to the loss of statistical power. Additionally, the trial was open label, which might have introduced the potential for ascertainment bias. Given that the trial involved four groups with different treatment principles, daily doses, and durations, the fabrication of a placebo would not have been feasible. In addition, there was consistency between the main outcome and the clinical efficacy among positive urine cultures, which helps mitigate concerns over bias. Women were followed up for only 1 month, which might have limited the detection of late-occurring events. However, this follow-up duration is consistent with that used in other studies of uncomplicated UTI that excluded patients with recurrent infections. We did not collect data on the specific racial or ethnic

backgrounds of participants; however, there was no exclusion of patients on the basis of racial or ethnic group, and some patients from minoritised groups contributed to the study design. Laboratory analyses were not centralised, which could have introduced variability in microbiological methods across different sites. However, this study was designed to be pragmatic, reflecting the real-world conditions of both health centres and microbiological departments. This also explains the high number of losses in the per-protocol population, as patients with persistent symptoms are more likely to contact a health-care provider than those who are clinically cured. The lack of microbiological data, particularly for patients with treatment failure who received an additional antibiotic course and were excluded from analysis, might have led to underestimation of differences between treatment groups. Consequently, microbiological cure rates appeared similar across all the treatment groups in our trial.

When selecting an antibiotic regimen, efficacy is paramount, but other factors, such as adherence and tolerability, should also be considered. Shorter regimens, such as one or two doses of fosfomycin, might improve adherence due to their simplicity, as shown in our study. However, overall adherence across all four antibiotic regimens was high, and patient satisfaction did not differ substantially between treatments. Adverse event rates were similar, although side-effect profiles varied: gastrointestinal symptoms were more common with fosfomycin; fatigue and somnolence with nitrofurantoin; and headache and nausea with pivmecillinam.

In conclusion, this pragmatic randomised clinical trial comparing first-line antibiotics for uncomplicated UTIs in women found that a single dose of fosfomycin had the lowest clinical effectiveness among the regimens tested, whereas nitrofurantoin had the highest effectiveness, followed by pivmecillinam, particularly in patients with positive baseline urine cultures. These findings suggest that the role of fosfomycin as a first-line treatment for uncomplicated UTIs should be re-evaluated.

Contributors

CL, RMon, AG-S, and RMor conceptualised and designed the study. The conduct of the trial and recruitment were performed by JM-C, RR-B, JMM, MAS-C, AL and AM. The microbiological analysis was coordinated by MA-S. The study drug management was performed by AT-M, and the pharmacovigilance was conducted by SF-G, CMJ, and RMor. CL, RMon, AG-S and RMor led the application for funding. The data analysis was performed by RMon and DO, with RMor directly accessing and verifying the underlying data reported in the manuscript. CL and RMon prepared the initial draft manuscript. All authors had full access to the data in the study and reviewed, edited, and approved the submission of the manuscript for publication.

Declaration of interests

JM-C has received payments from the Health Department of the Madrid Community and a research grant from the Carlos III Institute of Health, Ministry of Science and Innovation (Spain), under grant number ICI20/00148. SF-G has received grants from the Strategic Research and Innovation Plan for Health (Pla Estratègic de Recerca i Innovació en Salut) 2022–2024, funded by the Department of Health of the Government of Catalonia. RMor has participated in a Data Safety

Monitoring Board or Advisory Board for Archivel Farma and CONNECTA Therapeutics. All other authors declare no competing interests.

Data sharing

Individual, de-identified participant data used in these analyses can be made available, upon request to the corresponding author, to any qualified investigator following approval of a protocol and signed forms. The trial protocol, statistical analysis plan, and participant documents are available online at <https://www.idiapigol.org>.

Acknowledgments

We thank the general practitioners, nurses, patients, and the pharmacy and microbiology services staff who participated in the trial and contributed to advancing knowledge of antibiotic treatment for urinary tract infections. The study was funded by grants from the Carlos III Institute of Health, Ministry of Science and Innovation (Spain), awarded in the 2020 call under the Academic Clinical Trials Call, with reference IC120/00100, and by EU European Regional Development Fund. The funding was granted for a maximum duration of 4 years.

References

- Hooton TM. Clinical practice. Uncomplicated urinary tract infection. *N Engl J Med* 2012; **366**: 1028–37.
- Frimodt-Møller N, Bjerrum L. Treating urinary tract infections in the era of antibiotic resistance. *Expert Rev Anti Infect Ther* 2023; **21**: 1301–08.
- Nicolle LE. Uncomplicated urinary tract infection in adults including uncomplicated pyelonephritis. *Urol Clin North Am* 2008; **35**: 1–12.
- Castillo-Tokumori F, Irey-Salgado C, Málaga G. Worrisome high frequency of extended-spectrum beta-lactamase-producing *Escherichia coli* in community-acquired urinary tract infections: a case-control study. *Int J Infect Dis* 2017; **55**: 16–19.
- Bonkat G, Kranz J, Cai T, et al. EAU guidelines on urological infections. European Association of Urology, 2026. <https://d56bochluxqnx.cloudfront.net/documents/full-guideline/EAU-Guidelines-on-Urological-Infections-2026.pdf> (accessed April 3, 2026).
- Gupta K, Hooton TM, Naber KG, et al. International clinical practice guidelines for the treatment of acute uncomplicated cystitis and pyelonephritis in women: a 2010 update by the Infectious Diseases Society of America and the European Society for Microbiology and Infectious Diseases. *Clin Infect Dis* 2011; **52**: e103–20.
- Zollner-Schwetz I, König E. Treatment options for multidrug-resistant Gram-negatives in urinary tract infections. *Curr Opin Urol* 2023; **33**: 173–79.
- Llor C, Moragas A, Hernández S, Crispí S, Cots JM. Misconceptions of Spanish general practitioners' attitudes toward the management of urinary tract infections and asymptomatic bacteriuria: an internet-based questionnaire study. *Rev Esp Quimioter* 2017; **30**: 372–78.
- Verma T, Manhas GS, Manhas RS. Efficacy and safety of single-dose fosfomycin for uncomplicated urinary tract infection in women: systematic review and meta-analysis. *J Midlife Health* 2025; **16**: 124–36.
- Konwar M, Gogtay NJ, Ravi R, Thatte UM, Bose D. Evaluation of efficacy and safety of fosfomycin versus nitrofurantoin for the treatment of uncomplicated lower urinary tract infection (UTI) in women—a systematic review and meta-analysis. *J Chemother* 2022; **34**: 139–48.
- Huttner A, Kowalczyk A, Turjeman A, et al. Effect of 5-day nitrofurantoin vs single-dose fosfomycin on clinical resolution of uncomplicated lower urinary tract infection in women: a randomized clinical trial. *JAMA* 2018; **319**: 1781–89.
- Kaye KS, Santerre Henriksen A, Sommer M, Frimodt-Møller N. Safety and tolerability of pivmecillinam during more than four decades of clinical experience: a systematic review. *Clin Infect Dis* 2025; **80**: 280–99.
- Pinart M, Kranz J, Jensen K, et al. Optimal dosage and duration of pivmecillinam treatment for uncomplicated lower urinary tract infections: a systematic review and meta-analysis. *Int J Infect Dis* 2017; **58**: 96–109.
- Malmros K, Huttner BD, McNulty C, Rodríguez-Baño J, Pulcini C, Tängdén T, and the ESGAP UTI Working Group. Comparison of antibiotic treatment guidelines for urinary tract infections in 15 European countries: results of an online survey. *Int J Antimicrob Agents* 2019; **54**: 478–86.
- García-Sangenís A, Morros R, Aguilar-Sánchez M, et al, and the SCOUT Study Group. Clinical effectiveness and bacteriological eradication of three different short-course antibiotic regimens and single-dose fosfomycin for uncomplicated lower urinary tract infections in adult women (SCOUT study): study protocol for a randomised clinical trial. *BMJ Open* 2021; **11**: e055898.
- Hopewell S, Chan AW, Collins GS, et al. CONSORT 2025 statement: updated guideline for reporting randomised trials. *BMJ* 2025; **389**: e081123.
- Harris PA, Taylor R, Thielke R, Payne J, Gonzalez N, Conde JG. Research electronic data capture (REDCap)—a metadata-driven methodology and workflow process for providing translational research informatics support. *J Biomed Inform* 2009; **42**: 377–81.
- European Committee on Antimicrobial Susceptibility Testing (EUCAST). Breakpoint tables for interpretation of MICs and zone diameters, version 11.0, 2021. https://www.eucast.org/fileadmin/eucast/pdf/Document_Archive/bacteria/breakpoint_tables/v_11.0_Breakpoint_Tables.pdf (accessed April 3, 2026).
- European Medicines Agency. Guideline on the evaluation of medicinal products indicated for treatment of bacterial infections. May 19, 2022. https://www.ema.europa.eu/en/documents/scientific-guideline/guideline-evaluation-medicinal-products-indicated-treatment-bacterial-infections-revision-3_en.pdf (accessed Jan 16, 2026).
- Oduyayo A, Gryaznov D, Copsey B, et al, and the ASPIRE study group. Design, analysis and reporting of multi-arm trials and strategies to address multiple testing. *Int J Epidemiol* 2020; **49**: 968–78.
- Advani SD, Thaden JT, Perez R, Stair SL, Lee UJ, Siddiqui NY. State-of-the-art review: recurrent uncomplicated urinary tract infections in women. *Clin Infect Dis* 2025; **80**: e31–42.
- Kornfält Isberg H, Hedin K, Melander E, Mölstad S, Beckman A. Uncomplicated urinary tract infection in primary health care: presentation and clinical outcome. *Infect Dis (Lond)* 2021; **53**: 94–101.
- Wagenlehner F, Perry CR, Hooton TM, et al. Oral gepotidacin versus nitrofurantoin in patients with uncomplicated urinary tract infection (EAGLE-2 and EAGLE-3): two randomised, controlled, double-blind, double-dummy, phase 3, non-inferiority trials. *Lancet* 2024; **403**: 741–55.
- Butler CC, Francis N, Thomas-Jones E, et al. Variations in presentation, management, and patient outcomes of urinary tract infection: a prospective four-country primary care observational cohort study. *Br J Gen Pract* 2017; **67**: e830–41.
- Little P, Turner S, Rumsby K, et al. Validating the prediction of lower urinary tract infection in primary care: sensitivity and specificity of urinary dipsticks and clinical scores in women. *Br J Gen Pract* 2010; **60**: 495–500.
- Butler CC, Francis NA, Thomas-Jones E, et al. Point-of-care urine culture for managing urinary tract infection in primary care: a randomised controlled trial of clinical and cost-effectiveness. *Br J Gen Pract* 2018; **68**: e268–78.
- Lifshitz E, Kramer L. Outpatient urine culture: does collection technique matter? *Arch Intern Med* 2000; **160**: 2537–40.
- Badia JM, Calbo E, Besoli A, et al. Vigilància de les Infeccions relacionades amb l'atenció sanitària de Catalunya (VINCat): informe anual 2024. Barcelona: Servei Català de la Salut, 2025. <http://hdl.handle.net/11351/13925> (accessed Jan 16, 2026).
- Wijma RA, Koch BCP, van Gelder T, Mouton JW. High interindividual variability in urinary fosfomycin concentrations in healthy female volunteers. *Clin Microbiol Infect* 2018; **24**: 528–32.
- Michalopoulos AS, Livaditis IG, Gougoutas V. The revival of fosfomycin. *Int J Infect Dis* 2011; **15**: e732–39.
- Fedrigio NH, Mazucheli J, Albiero J, et al. Pharmacodynamic evaluation of fosfomycin against *Escherichia coli* and *Klebsiella* spp. from urinary tract infections and the influence of pH on fosfomycin activities. *Antimicrob Agents Chemother* 2017; **61**: e02498–16.